遙測與空間資訊之分析與應用

國家災害防救科技中心 資訊組
台灣大學土木工程學系 教授
蘇文瑞
wraysu@ntu.edu.tw

Course Overview & Philosophy

"Learning by Exploring, Creating by Thinking"

- Shift in Paradigm:

  From "How to write code" to "What to build with AI."
- Vibe Coding:

  Leveraging AI Agents (Windsurf/Cursor) to focus on Intent and Logic rather than just syntax.
- Problem-Oriented:

  Real-world disaster prevention cases as the driving force.
- AI as a Co-pilot:

  AI is your assistant, but You are the Captain responsible for verification.

Class attendance and grading method

Grading Method

- **Exercises & Homework (30%)**: Weekly Vibe Coding tasks + AI interaction logs.
- **Mid-term Proposal (30%)**: Project definition and technical feasibility.
- **Final Project (40%)**: A functional spatial AI application (e.g., Warning System).

Roadmap

- **Weeks 1-4:** Foundations & Professional Standards.
- **Weeks 5-8:** Big Data & Real-time Dashboards (Streamlit).
- **Weeks 9-12:** Satellite Imagery (STAC/Sentinel-2) & Machine Learning.
- **Weeks 13-16:** Final Project Sprint & Innovation Showcase.

Learning Objectives

- Master AI-Assisted Development: Design effective prompts and manage AI-human collaboration.
- Professional Infrastructure: Manage projects using .env, requirements.txt, and GitHub.
- Spatial Problem Solving: Solve unique challenges in geospatial data using Python.
- Disaster Innovation: Develop real-time monitoring and analysis tools for disaster prevention.

Why Python for GIS?

The Power of Open Source & Free Tools

- **Zero Licensing Fees:**
  Unlike ArcGIS, Python libraries (GeoPandas, Leafmap) are **Free** for life.
- **AI IDE Accessibility:**
  * **Windsurf/Cursor:** Provide a **Hobby (Free) Tier** sufficient for classroom exercises.
- **No "Pay-to-Learn":**
  Students can start innovating without financial barriers.
- **Efficiency & Scale:**
  Optimized for **Big Data** and satellite imagery.
- **Open Community:**
  Reproducible research that anyone can verify.

Research Case 1: Quantitative Correlation

Groundwater Level vs. Land Subsidence

- **Spatial Problem:** Correlating discrete monitoring well data (Vector) with continuous satellite imagery (Raster).
- **Research Logic:**
  - Spatio-temporal data cleaning (AI assisted).
  - Statistical correlation analysis using PySAL.
  - Visualizing subsidence hot-spots for policy making.
- **AI Task:** Automate the multi-source data merging and trend detection.

Research Case 2: Decision Support Modeling

Real-time Logistics for Disaster Response

- **Spatial Problem:** Optimizing chemical tanker routes during extreme weather (Rainfall + Flooding).
- **Research Logic:**
  - Dynamic Road Network Analysis (NetworkX).
  - Integration of Real-time Rainfall APIs.
  - Evacuation path optimization under shifting constraints.
- **AI Task:** Build a real-time routing engine that reacts to live weather feeds.

Research Case 3: Land Cover Change Detection

Multi-spectral Analysis for Disaster Assessment

- **Scenario:** Landslide detection after an earthquake or Burn severity after a forest fire.
- **Research Logic:**
  - **Pre-event vs. Post-event:** Comparing spectral indices (NDVI/NBR).
  - **Spectral Math:** Calculating Change Indices.
  - **Classification:** Segmenting impacted areas based on statistical thresholds.
- **AI Task:** Automated STAC retrieval, cloud masking, and multi-band arithmetic.

Research Case 4: Coastal "Green Lungs" (Mangroves)

Monitoring Carbon Absorption Capacity (New Simplified Case!)

- **Scenario:** "How much CO2 is our coastal forest breathing in?"
- **Research Logic:**
  - **Measure the "Lungs" (Biomass):** Use Satellite Radar (SAR) to see through the clouds and measure how "thick" and "tall" the mangrove forest is.
  - **Check the "Breathing Rate" (Flux):** Correlate Satellite Greenness (NDVI/EVI) with local sensors to see how fast the forest is absorbing carbon.
  - **Trend Analysis:** Is our "Green Lung" growing healthier or shrinking due to pollution/sea-level rise?
- **AI Task:** AI helps bridge the gap between "Satellite Photos" and "Carbon Numbers" using regression models.

What Python for GIS can do?

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029740 99999 195201010000 200  23 ***  15 OVC 7 2 *  5.0 63 ** ** ** ** ** ** ** 6
029740 99999 195201010600 220  18 ***   8 OVC 7 2 *  2.2 63 ** ** ** ** ** ** ** 6
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029740 99999 195201061800 130  18 ***  26 BKN 5 * * 12.5 02 ** ** *
029740 99999 195201070000 200  14 ***   8 OVC 7 * *  1.9 51 ** ** *
029740 99999 195201070600 250   2 ***   2 OVC 7 * *  1.5 51 ** ** *
029740 99999 195201071200 250  18 ***   5 OVC 5 * *  6.2 02 ** ** *
029740 99999 195201071800 270  17 ***  15 OVC 7 * *  3.8 59 ** ** *
029740 99999 195201080000 290  16 *** 722 CLR 0 0 0 12.5 00 ** ** *
...
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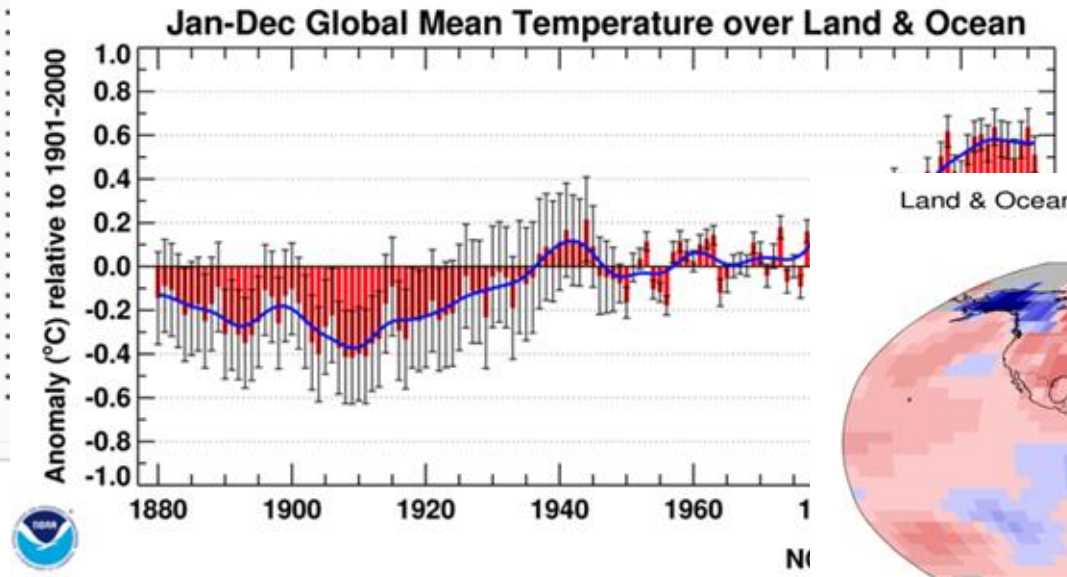
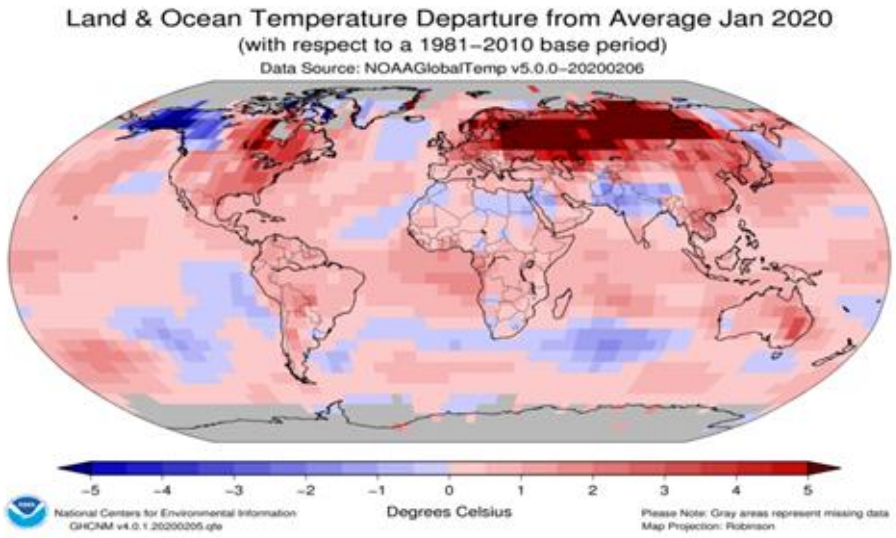
Big Data

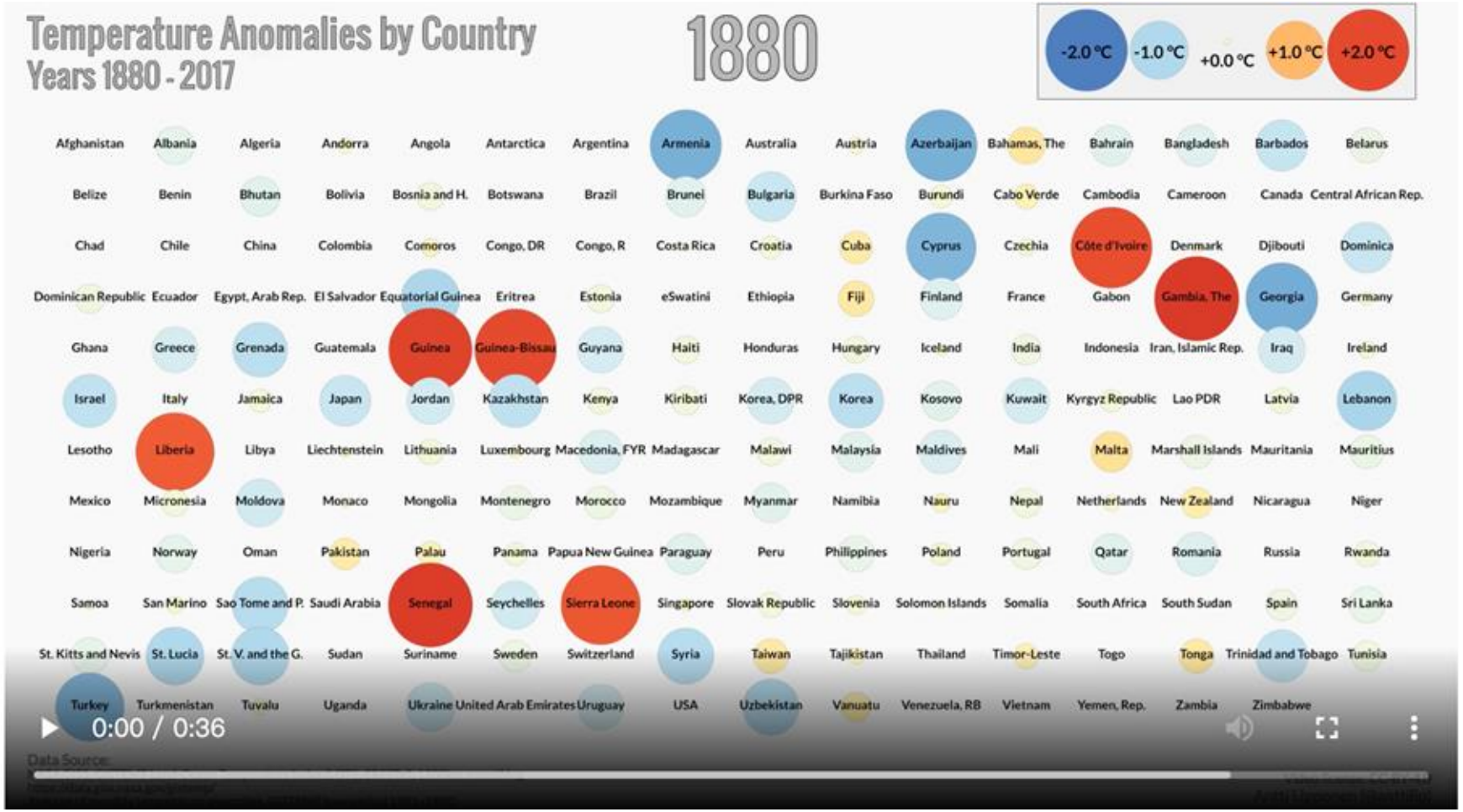


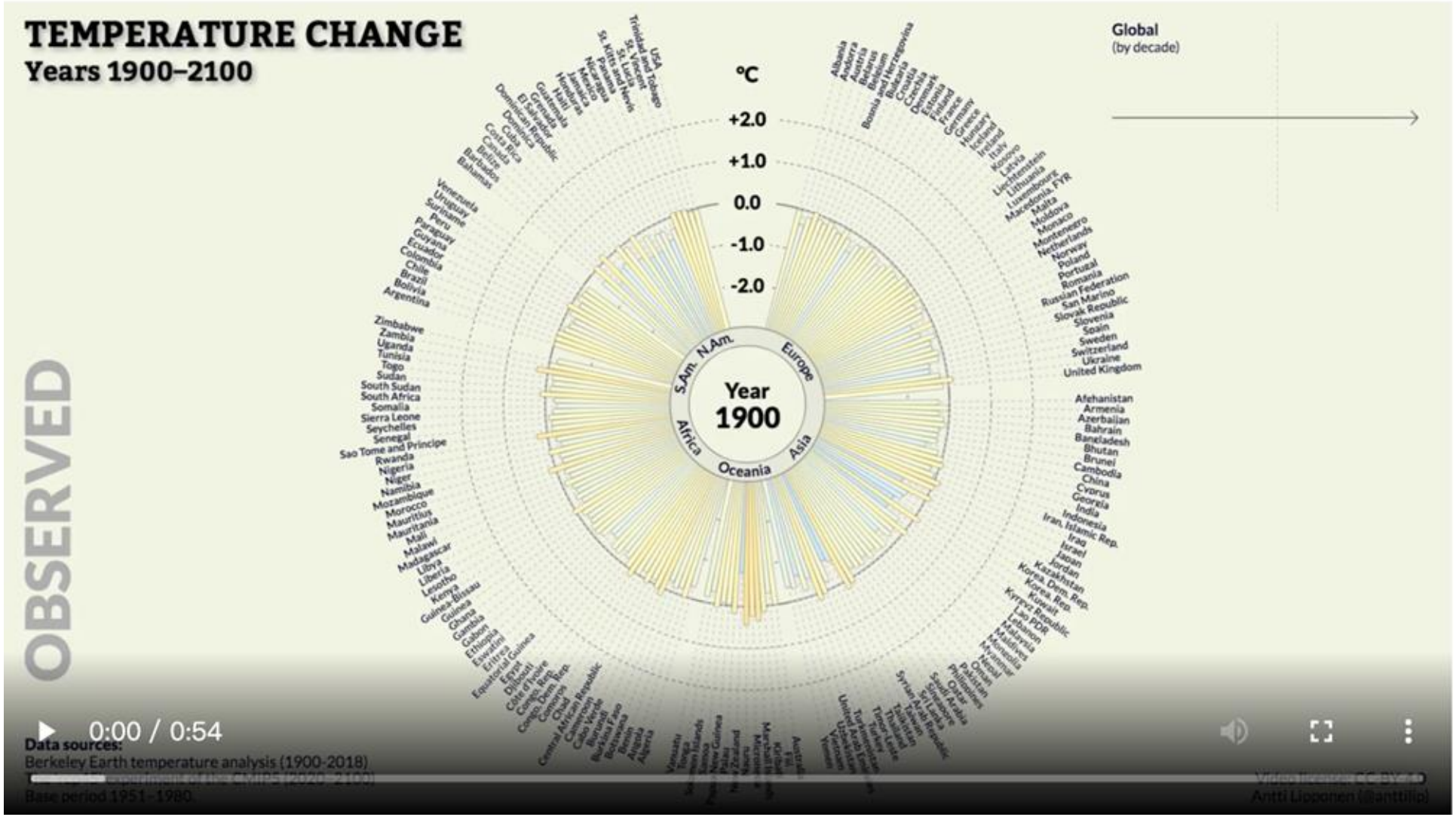
Analyzing



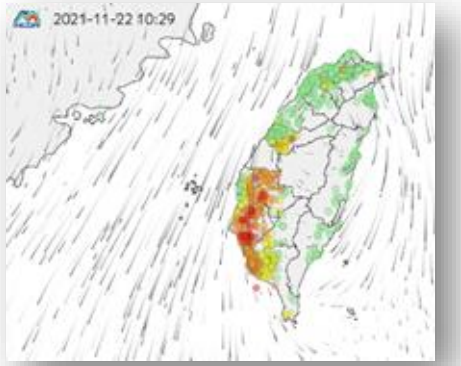
Global temperature anomalies for January 2020. Source:
https://www.ncdc.noaa.gov/sotc/global/201603

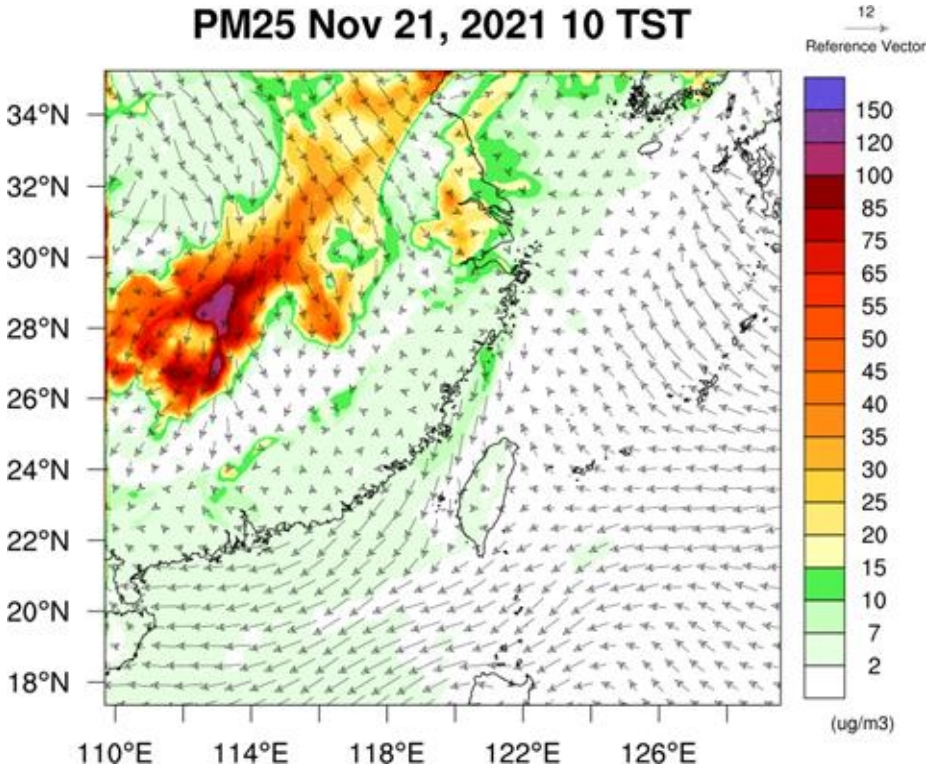
Visualizing

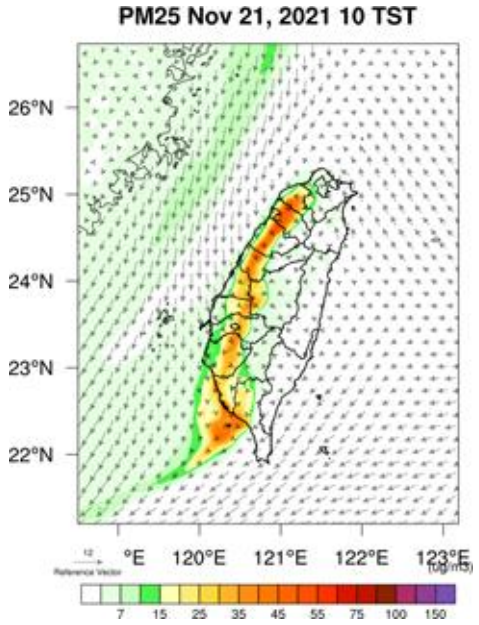

Temperature Anomalies by Country
Years 1880 - 2017
1880
-2.0 °C
-1.0 °C
+0.0 °C
+1.0 °C
+2.0 °C
Afghanistan Albania Algeria Andorra Angola Antarctica Argentina Armenia Australia Austria Azerbaijan Bahamas, The Bahrain Bangladesh Barbados Belarus
Belize Benin Bhutan Bolivia Bosnia and H. Botswana Brazil Brunei Bulgaria Burkina Faso Burundi Cabo Verde Cambodia Cameroon Canada Central African Rep.
Chad Chile China Colombia Comoros Congo, DR Congo, R Costa Rica Croatia Cuba Cyprus Czechia Côte d'Ivoire Denmark Djibouti Dominica
Dominican Republic Ecuador Egypt, Arab Rep. El Salvador Equatorial Guinea Eritrea Estonia eSwatini Ethiopia Fiji Finland France Gabon Gambia, The Georgia Germany
Ghana Greece Grenada Guatemala Guinea Guinea-Bissau Guyana Haiti Honduras Hungary Iceland India Indonesia Iran, Islamic Rep. Iraq Ireland
Israel Italy Jamaica Japan Jordan Kazakhstan Kenya Kiribati Korea, DPR Korea Kosovo Kuwait Kyrgyz Republic Lao PDR Latvia Lebanon
Lesotho Liberia Libya Liechtenstein Lithuania Luxembourg Macedonia, FYR Madagascar Malawi Malaysia Maldives Mali Malta Marshall Islands Mauritania Mauritius
Mexico Micronesia Moldova Monaco Mongolia Montenegro Morocco Mozambique Myanmar Namibia Nauru Nepal Netherlands New Zealand Nicaragua Niger
Nigeria Norway Oman Pakistan Palau Panama Papua New Guinea Paraguay Peru Philippines Poland Portugal Qatar Romania Russia Rwanda
Samoa San Marino Sao Tome and P. Saudi Arabia Senegal Seychelles Sierra Leone Singapore Slovak Republic Slovenia Solomon Islands Somalia South Africa South Sudan Spain Sri Lanka
St. Kitts and Nevis St. Lucia St. V. and the G. Sudan Suriname Sweden Switzerland Syria Taiwan Tajikistan Thailand Timor-Leste Togo Tonga Trinidad and Tobago Tunisia
Turkey Turkmenistan Tuvalu Uganda Ukraine United Arab Emirates Uruguay USA Uzbekistan Vanuatu Venezuela, RB Vietnam Yemen, Rep. Zambia Zimbabwe
0:00 / 0:36
Data Source:


TEMPERATURE CHANGE
Years 1900–2100
OBSERVED
°C
+2.0
+1.0
0.0
-1.0
-2.0
Year
1900
N.Am.
Europe
Asia
Oceania
Africa
S.Am.
Global
(by decade)
0:00 / 0:54
Data sources:
Berkeley Earth temperature analysis (1900-2018)
Base period 1951-1980

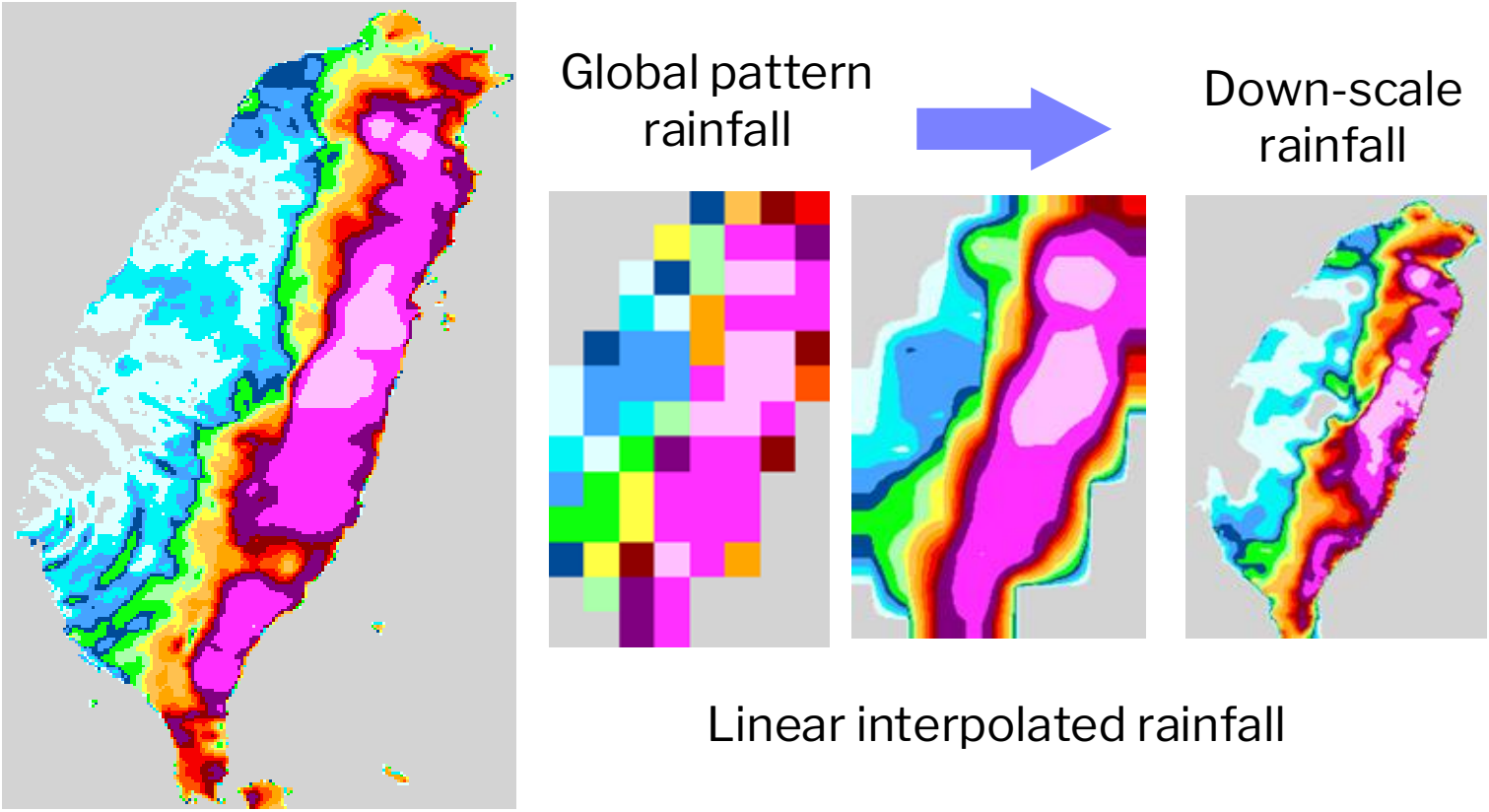
Visualizing Big Data







Analyzing Big Data



The Vibe Coding Manifesto (Dos & Don'ts)

✅ The "Dos"

- ✓ **Verify Geometric Integrity:** AI might ignore CRS (Coordinate Reference Systems). **Always check your EPSG.**
- ✓ **Document the Logic:** Use AI to write **Traditional Chinese comments** explaining the "Why," not just the "How."
- ✓ **Modularize Research:** Break your analysis into "Data Fetch -> Analysis Logic -> Scientific Output."

❌ The "Don'ts"

- × **Don't Hardcode Secrets:** Never put API Keys in scripts. Use .env.
- × **Don't Blindly Trust ML Models:** An AI-generated model is only as good as the spatial data you feed it.
- × **Don't Commit Sensitive Data:** Use .gitignore to keep raw data and private keys out of GitHub.

IDE: Your AI Agent Workspace

- **Windsurf & the "Cascade" Agent**
- **IDE: Integrated Development Environment**. It's your research laboratory.
- **Windsurf Agent:** Unlike a simple chatbot, Windsurf can see your files and run terminal commands.
- **Cascade Mode (The Brain):**
  - Command + L: Discuss research logic with AI.
  - Command + I: Ask AI to modify specific lines of code.
- **Contextual Awareness:** AI knows your project structure. Mention files with @filename.

CLI: The Developer's Engine Room

- **What is CLI (Command Line Interface)?**
- **CLI:** A text-based interface for precise control over your system.
- **Key Functions for Researchers:**
  - **Environment Management:** pip install to manage libraries.
  - **Version Control:** git and gh for syncing research to the cloud.
  - **Gemini CLI:** Use AI directly in the terminal for quick code reviews or logic brainstorming.

GitHub: Your Research Repository

Version History is your "Time Machine"

- Why GitHub for Researchers?
  1. Proof of original work (Commit history).
  2. Collaboration with other researchers.
  3. Professional portfolio for your career.
- The Workflow:
  1. git init: Start tracking.
  2. git add . -> git commit: Snapshot your progress.
  3. git push: Sync with GitHub servers.

Professional Standards I: Infrastructure

Secure Environment & Reproducibility

- 🔑 **API Keys (.env):** * Keep your Central Weather Admin (CWA) keys in .env.
  - CWA_API_KEY=your_secret_key
- 📦 **Dependencies (requirements.txt):**
  - pip freeze > requirements.txt.
  - Ensure your peers can replicate your analysis environment perfectly.

Professional Standards II: Directory Structure

A Clean Lab leads to Clear Results

```
/my_spatial_research
├── .env              # Private API Keys (DO NOT PUSH)
├── .gitignore        # Rules for Git (Exclude .env, /data)
├── requirements.txt  # Library list
├── main.py           # Main analysis logic
├── /scripts          # Helper functions
├── /data             # Raw research data (local only)
└── /outputs          # Resulting GeoJSON, CSV, and Maps
```

Dealing with Errors

"Error Messages are Research Guidance"

- **Step 1: Don't Panic.** Red text in the terminal is data, not failure.
- **Step 2: The Loop.**
  - Copy the full error.
  - Paste into **Cascade/Agent**.
  - Ask: "Explain the spatial logic error and suggest a fix for my CRS."
- **Step 3: Verification.** Ask AI how to write a "Test" to ensure the error is gone.

Advanced Visualization: Analysis over Art

- **Visualizing for Decision Making:** We use maps to find **patterns**, not just to look pretty.
- **Tools:** folium, leafmap, matplotlib, plotly.
- **Vibe Strategy:** Ask AI to "Create a map that highlights only the areas where the Z-score is above 1.96 (95% confidence)."

Remote Sensing Intro: STAC API

Accessing Global Data on Demand

- **The Problem:** Satellite scenes are massive (GBs).
- **The Solution: STAC** (SpatioTemporal Asset Catalog).
- **The Benefit:** Stream only the pixels you need for your analysis area.
- **Vibe Strategy:** AI writes the query; you focus on the **Spectral Analysis** (NDVI, NDWI).

Today's Mission (Part A): Infrastructure

"Setting up your Research Lab"

- Open **Windsurf/Cursor**.
- **Intent:** Ask AI to "Initialize a professional GIS project structure including .env and .gitignore."
- **API Setup:** Put your MOENV API Key into .env.
- **Git Setup:** Initialize your repo and link it to your GitHub account using gh auth login.

Today's Mission (Part B): Analysis

"Real-time AQI Analysis"

- **Intent:** Ask AI to "Fetch real-time AQI data from MOENV (aqx_p_432) and identify stations with AQI > 50."
- **Geospatial Logic:** "Calculate the distance from each high-AQI station to Taipei Main Station (25.0478, 121.5170)."
- **Output:** Generate an interactive map with color coding (Green/Yellow/Red) and export the list as a CSV in /outputs.
- **Publish:** git add . -> git commit -> git push.

Grading & Quota Management

- **AI Quota:** Use your **Windsurf Hobby (Free)** messages strategically.
- **Fallback:** For architectural questions, use **Gemini CLI** or the web version of Claude/ChatGPT.
- **Verification Grade:** Your commit messages must describe **what analysis** you performed.

Q&A & Resource Links

- **MOENV Open Data:** https://data.moenv.gov.tw/
- **Tool Docs:** Windsurf Cascade Guide
- **Student Benefits:** GitHub Student Developer Pack
- **API Access:** Google AI Studio (Gemini API)

What Will You Gain from This Course?

- **Learn Python for GIS & AI applications**
- **Develop a project with real-world applications**
- **Improve future job competitiveness (GIS/AI data analysis skills)**